

# Unlocking Day2Ops Excellence

**CISCO** Live !

Nexus Dashboard Innovations for ACI & NX-OS N9K Fabrics

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# Webex App

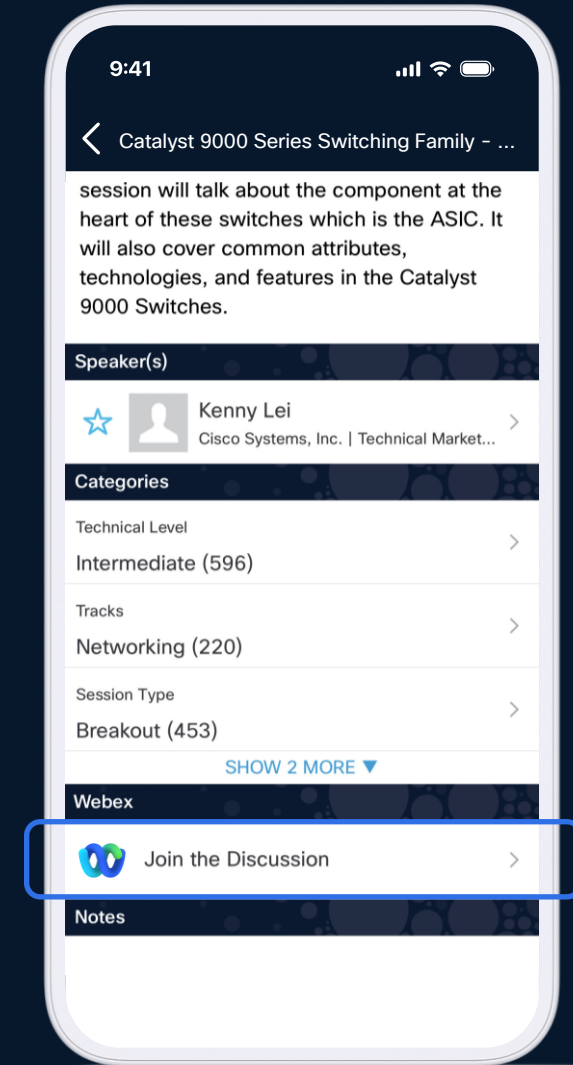
## Questions?

Use Webex App to chat with the speaker after the session

## How

- 1 Find this session in the Cisco Events App
- 2 Click “Join the Discussion”
- 3 Install the Webex App or go directly to the Webex space
- 4 Enter messages/questions in the Webex space

**Webex spaces will be moderated by the speaker until February 27, 2026.**



# Agenda



Day 2 Ops



Nexus Dashboard  
Overview



Features



Troubleshooting Demos



Conclusion

# Day 2 OPS



# Managing the Complete Lifecycle

Day  
**0**

**Infrastructure  
planning,  
delivery and  
installation**

Day  
**1**

**Discover,  
configure  
and secure**

Day  
**2**

**Assure,  
monitor and  
troubleshoot**

"The average cost of IT downtime is \$5,600 per minute" – Gartner



"Organizations that focus on proactive problem management can resolve issues 43% faster compared to reactive problem solving." – IDC (International Data Corporation)



# What's happening on Day 2 and what needs to change?



# Customer Challenges



Tools

## Fragmented

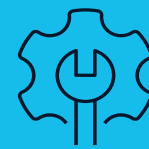
- Too many tools addressing silo-ed visibility use cases
- Some are old, and some are expensive
- Different protocols/mechanisms



Analyze

## Reactive

- Inconsistent API architecture
- Specialized knowledge required
- Difficult to root-cause issue, and often time too late to react to it



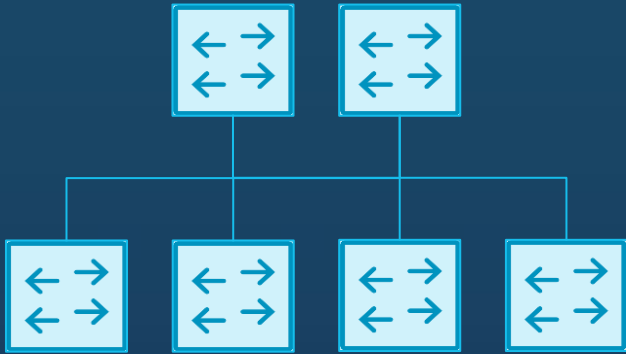
Insights

## Limited

- Low data fidelity that is not actionable
- Lack of data correlation, don't get the full picture
- No dataplane visibility

# Network Telemetry Frees the Data

Where Data Is Created



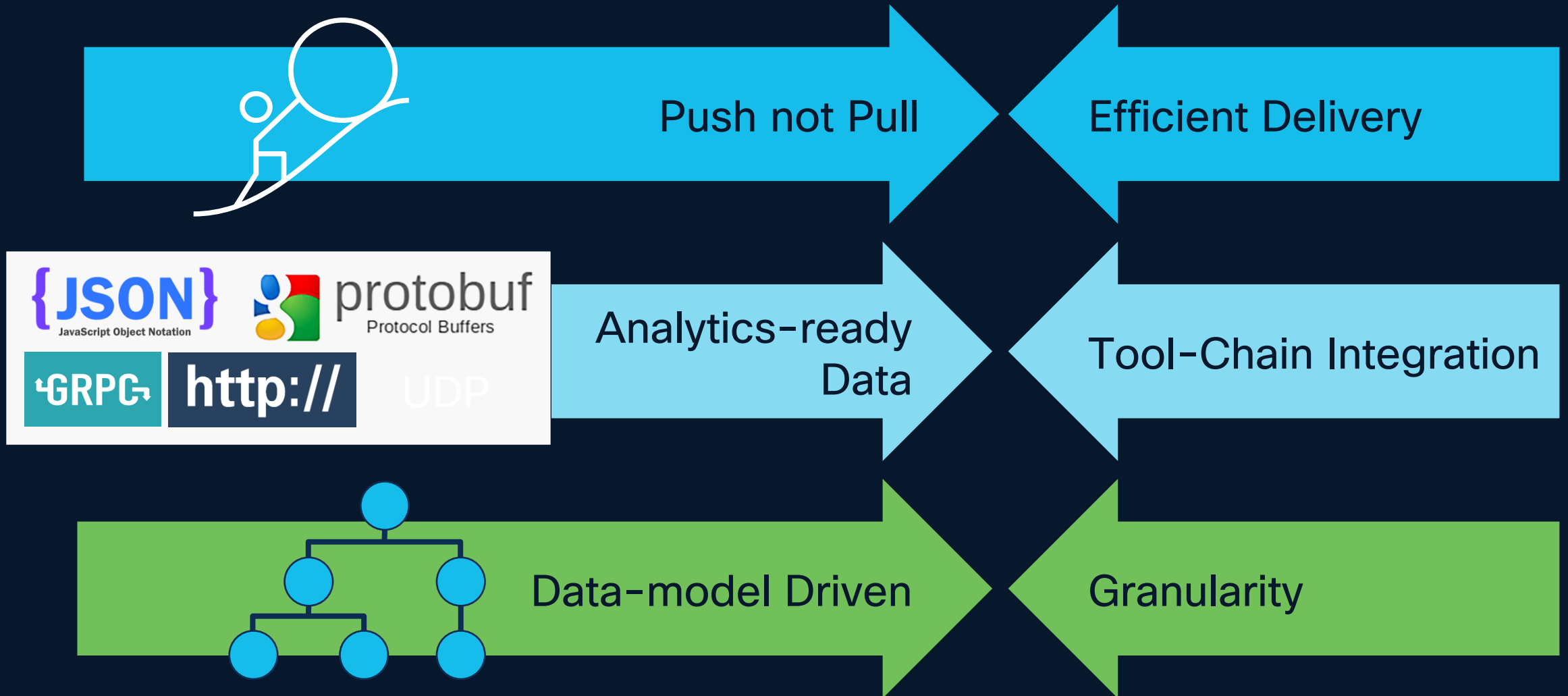
**As Much Useful Data  
As Efficiently as Possible**

Where Data Is Useful



Storage &  
analysis

# Key Streaming Telemetry Characteristics



# Nexus Dashboard Overview

# Nexus Dashboard

Operations Use cases and highlights



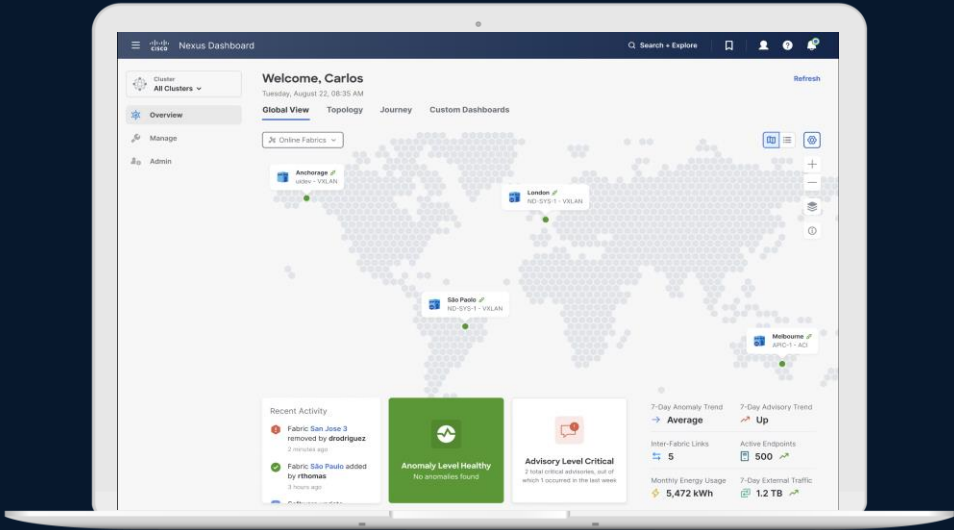
Deploy

Manage

Analyze

Security and Segmentation

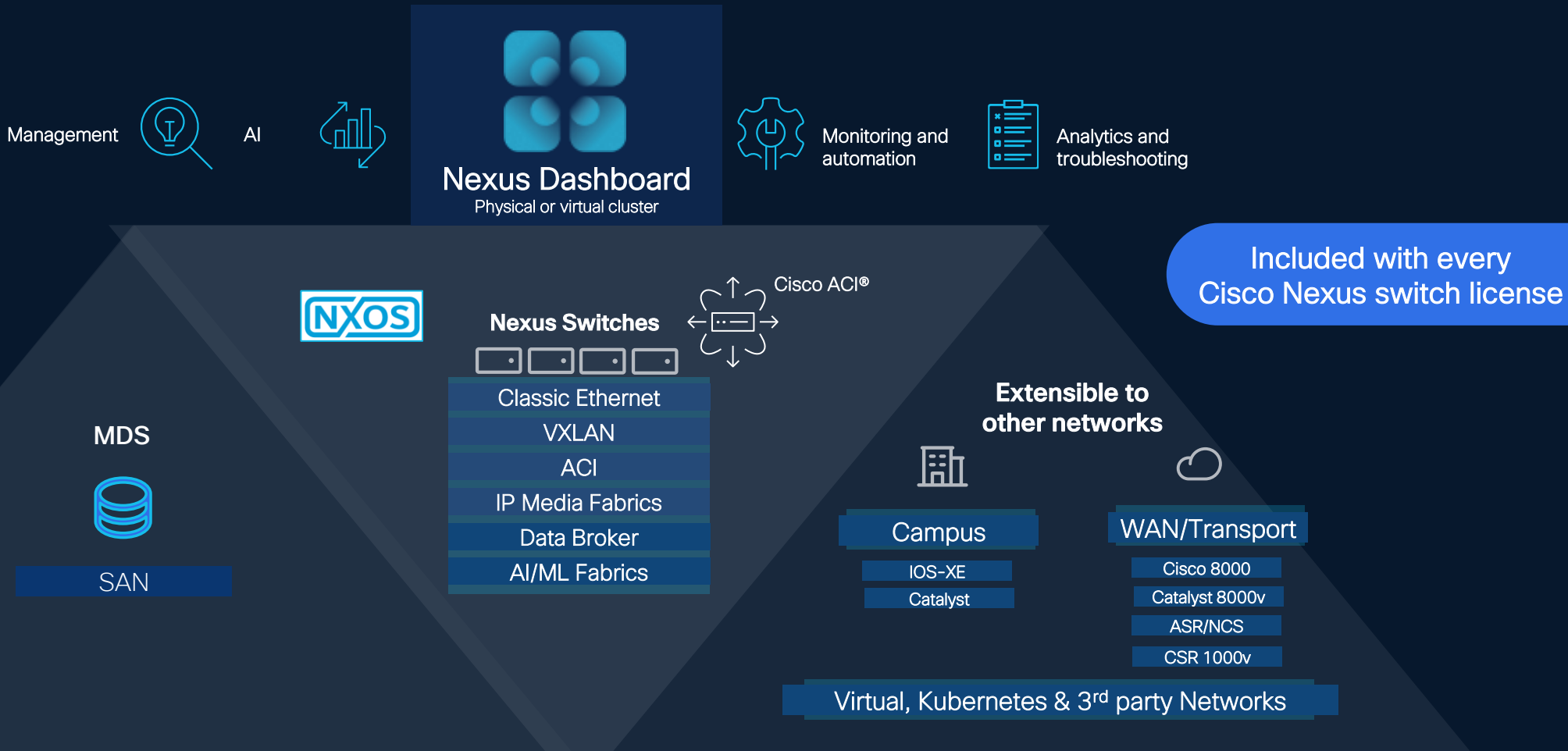
Interoperability





# Unified data-center network operations

Visibility, analytics, troubleshooting tools, and compliance all in one place



The simplest way to manage and operate data center connectivity\*

# Nexus Dashboard

## Nexus Dashboard Form Factors and scale



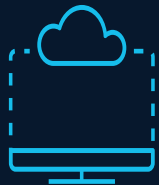
One Node pND  
Up to 50 switches



Three Node pND  
Up to 500 switches



Six Node pND  
Up to 1000 switches



One Node vND Data  
Up to 25 switches



Three Node vND  
Up to 100 switches



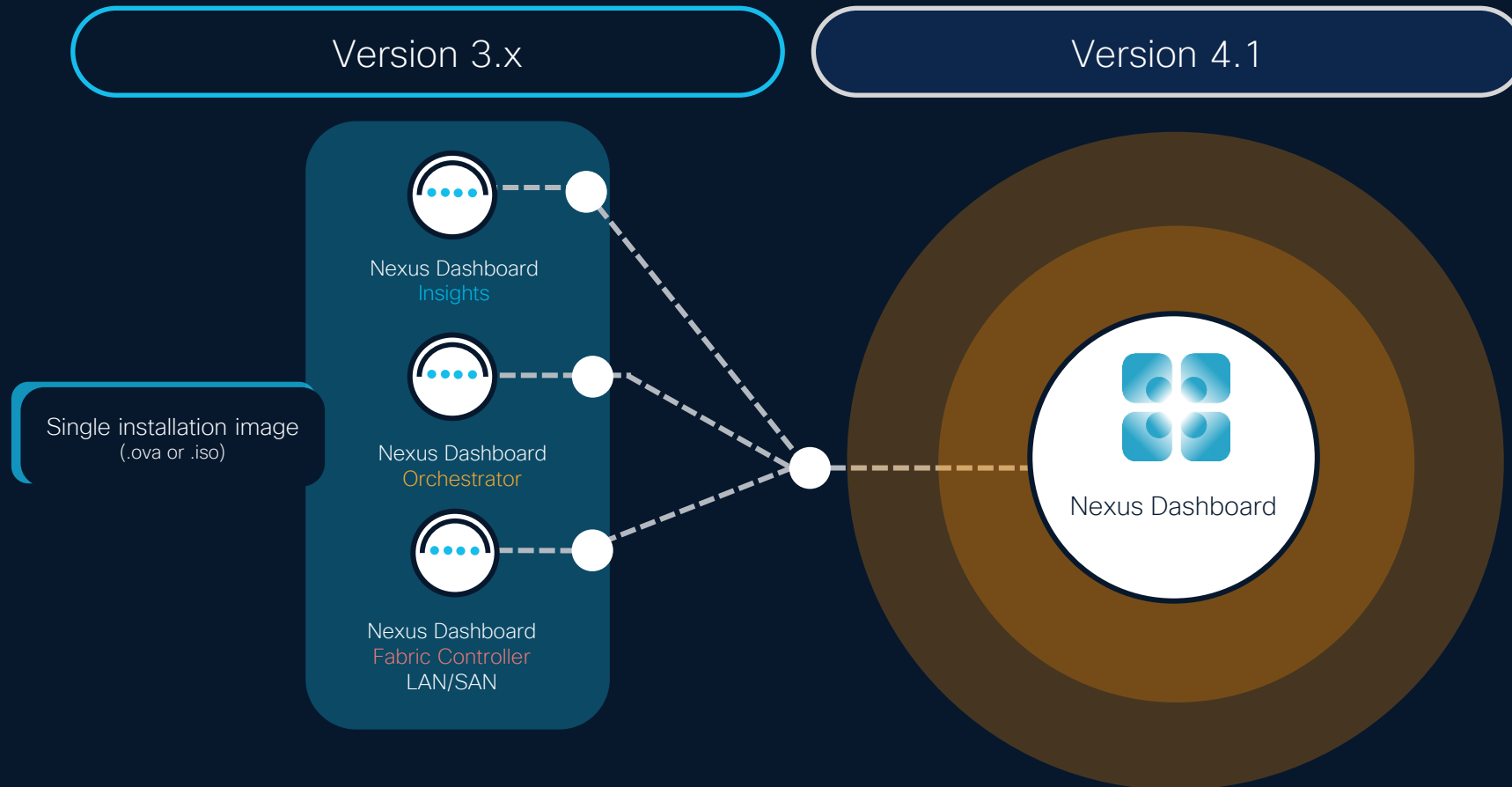
Six Node vND  
Up to 200 switches

# Public cloud Availability



# Unified Nexus Dashboard

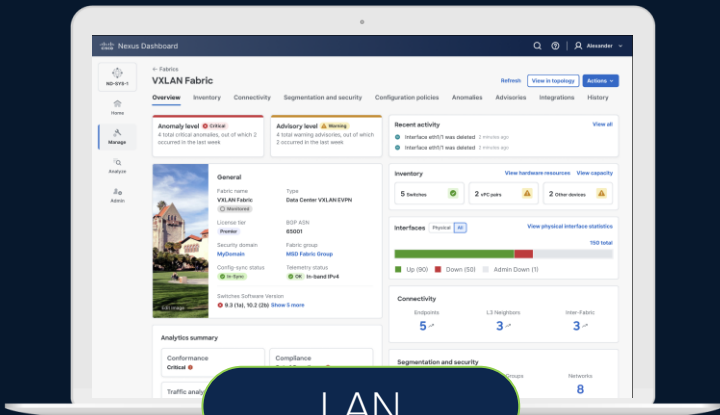
Toward simplification and unification



# Nexus Dashboard

## Deployment Modes

### LAN Persona

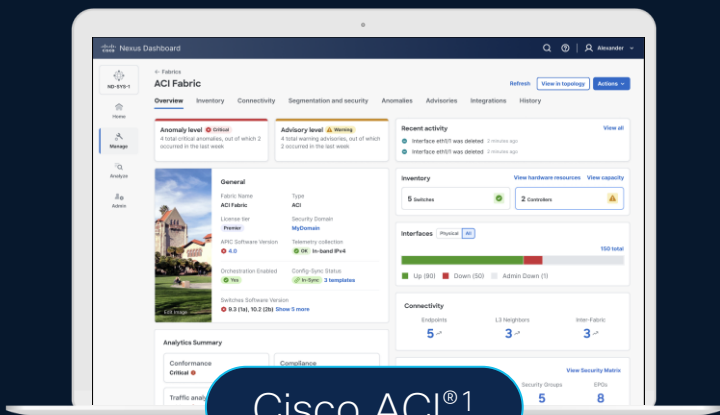


LAN

Zero-Touch classic, VXLAN, IPFM, Data Broker, AI/ML Networks & more

Nexus® 9000, Cisco Catalyst® 9000, IOS-XR and third party

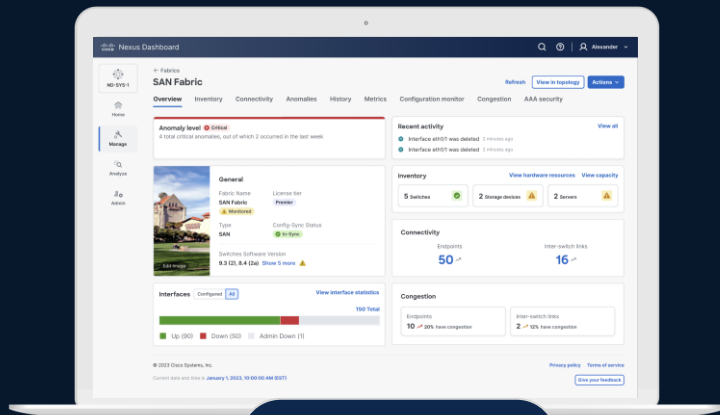
Cisco ACI®<sup>1</sup>



Zero-Touch VXLAN network with policy-oriented model

Nexus 9000

### SAN Persona



SAN

Zero-Touch customizable templates and Smart Zoning

Cisco MDS 9000

Consistent operational experience

Open APIs  
with Ansible, Terraform

Operate multiple switches  
as one logical unit



# Nexus Dashboard – Green/Brown Field

LAN Fabric automation and/or onboarding for AI analytics

Classic

2 or 3-tier NX-OS architectures

VXLAN

BGP EVPN fabrics for Nexus & Catalyst (IOS-XE)

AI/ML

Fine-tuned network provisioning to transport AI/ML apps with the best performance

External & Inter-Fabric Networks

NX-OS, IOS-XE, XR & 3<sup>rd</sup> party networks mainly used to interconnect ACI, NX-OS, & Campus

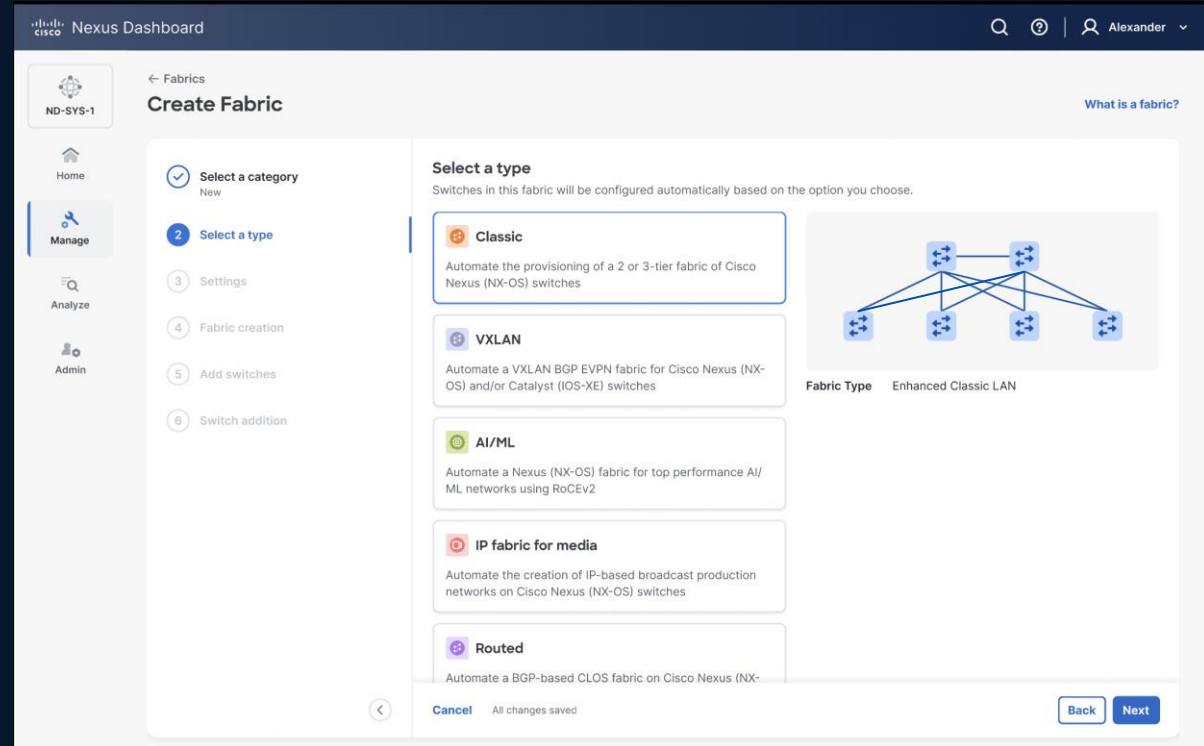
IP fabrics for media

Fine-tuned network provisioning and monitoring for broadcasting and media

Routed

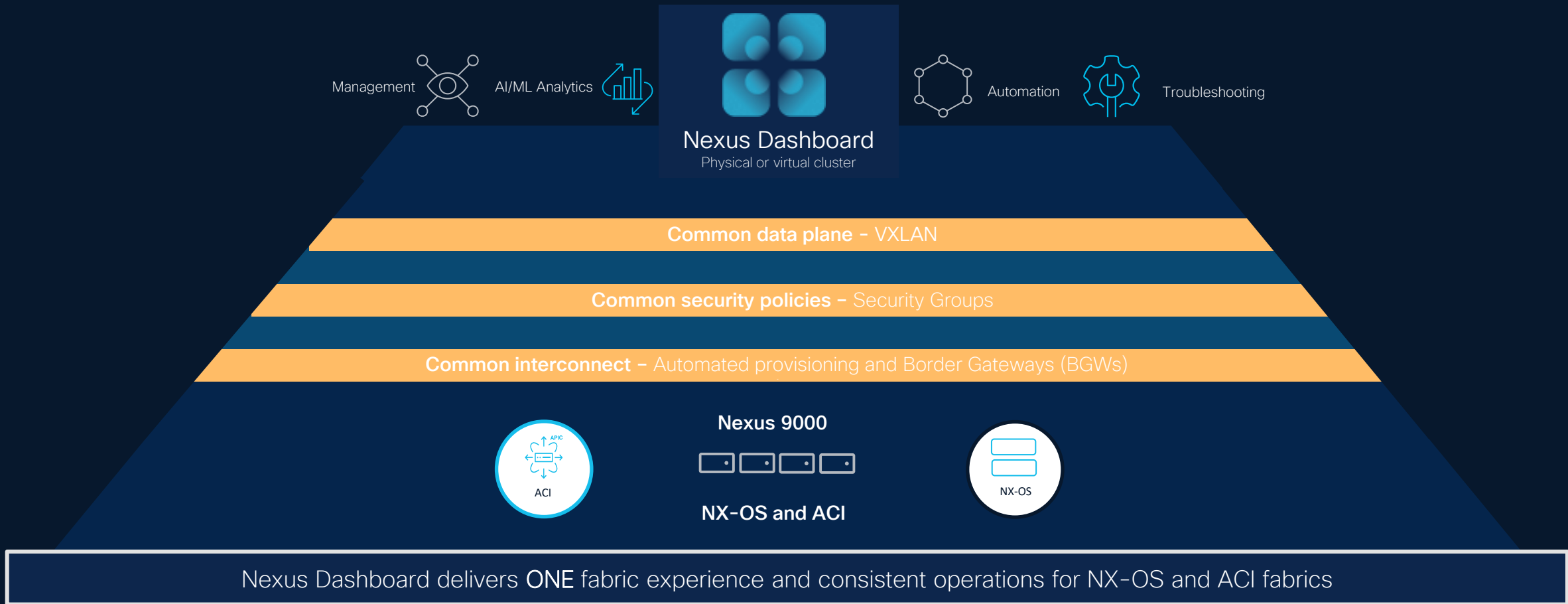
BGP-based CLOS NX-OS fabrics

ACI



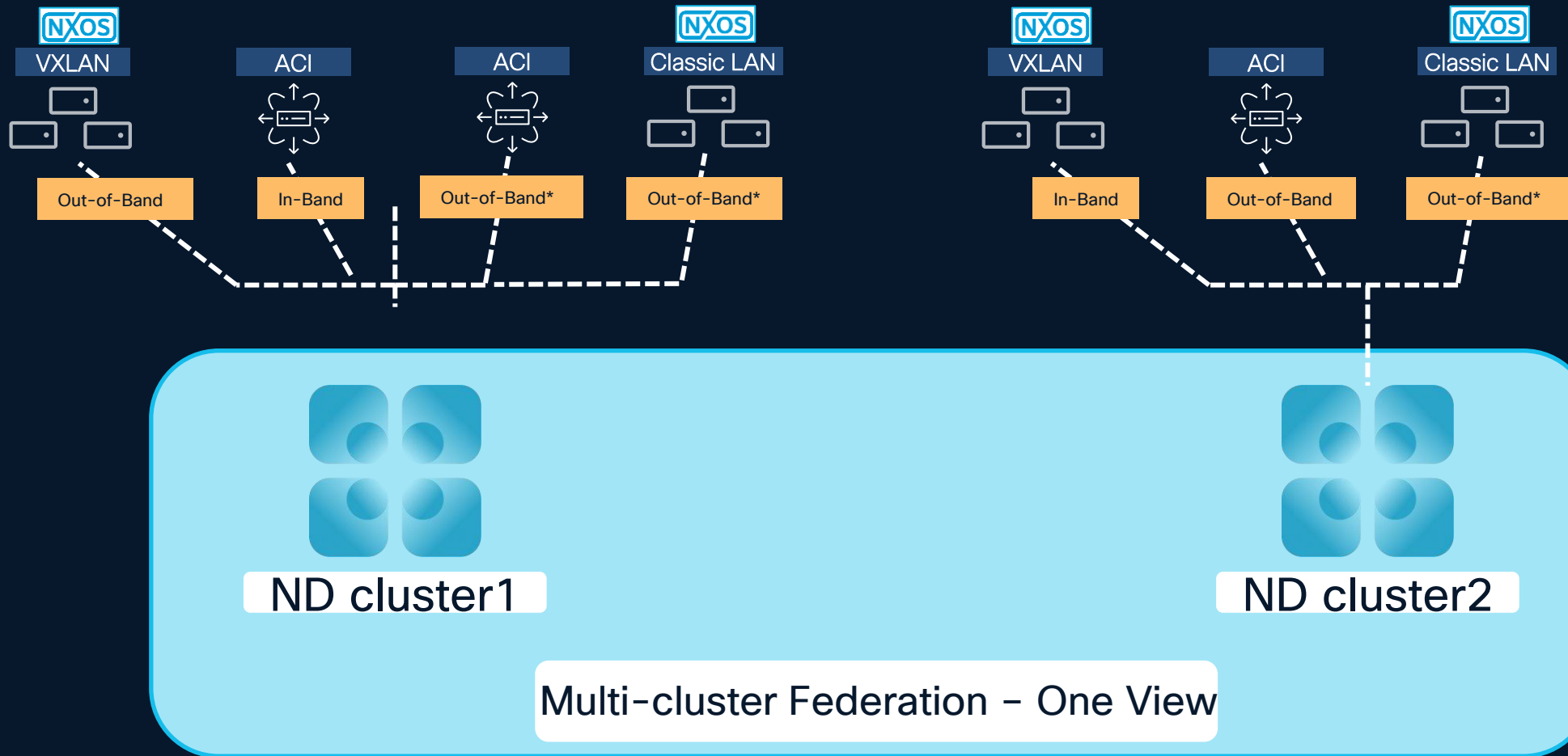
LAN

# The power of ONE Nexus fabric experience



# Unified Nexus Dashboard

Unify the operational experience



\*ACI OOB support as from ND 4.1 and ACI 6.1.2f

\*Classic LAN only supports OOB – Inband planned for ND 4.2

# Introducing Cisco Nexus Open Networking (ONe)

ND 4.1

## Solution Components

Nexus Dashboard as single point of control and operations

Consistent and open fabric experience across platforms

Nexus Dashboard



Inter-op

Inter-op\*

NX-OS VXLAN EVPN

Cisco ACI

Cisco ACI

Nexus 9000

Nexus 9000

Nexus 9000

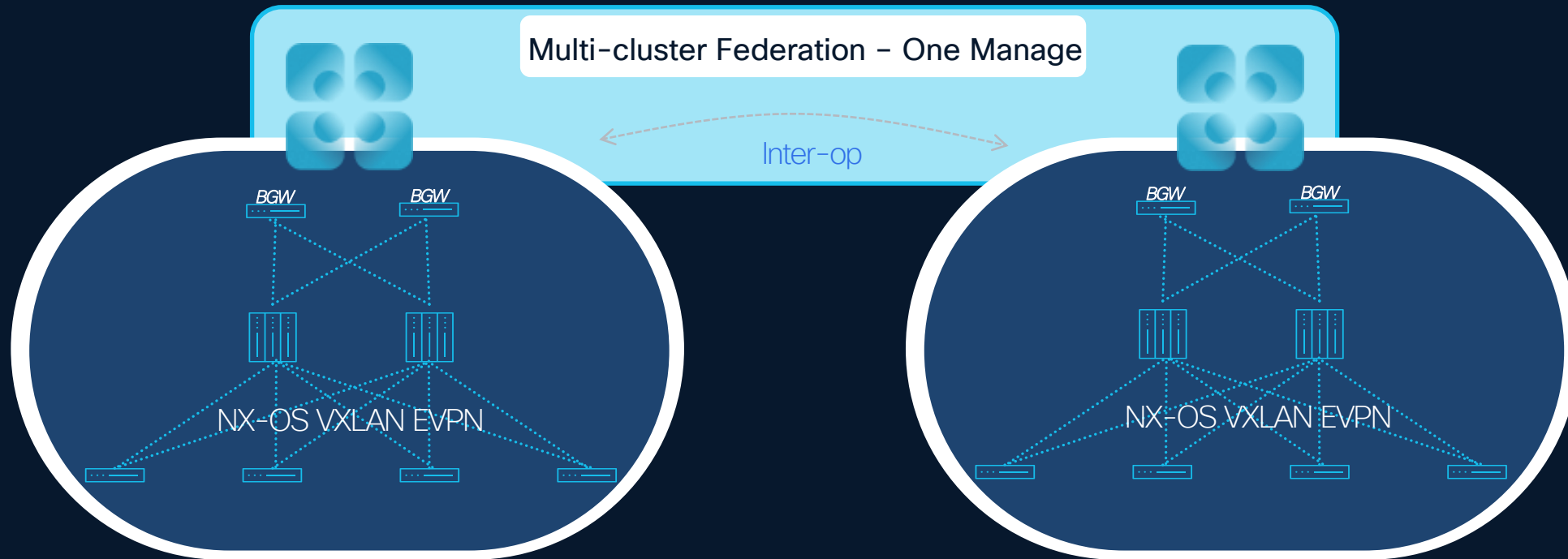
Common policy

- Zero trust networking and micro-segmentation based on endpoint / security groups
- Advanced service chaining and service redirection
- Administrative multi-tenancy
- Standards based inter-op with 3<sup>rd</sup> party networks using open networking protocols
- Integration with CI/CD pipelines with DevOps ready APIs

# ND One Manage

ND 4.1

- Customers can migrate their existing Nexus Dashboard Orchestrator (NDO) managed NX-OS VXLAN BGP EVPN Multi-Site environment to a multi-cluster fabric group that is driven from the ND One Manage view.



**Deploy and manage VXLAN Group Policy Options (GPOs) across multiple VXLAN BGP EVPN fabrics and clusters ensuring policy consistency at scale.**

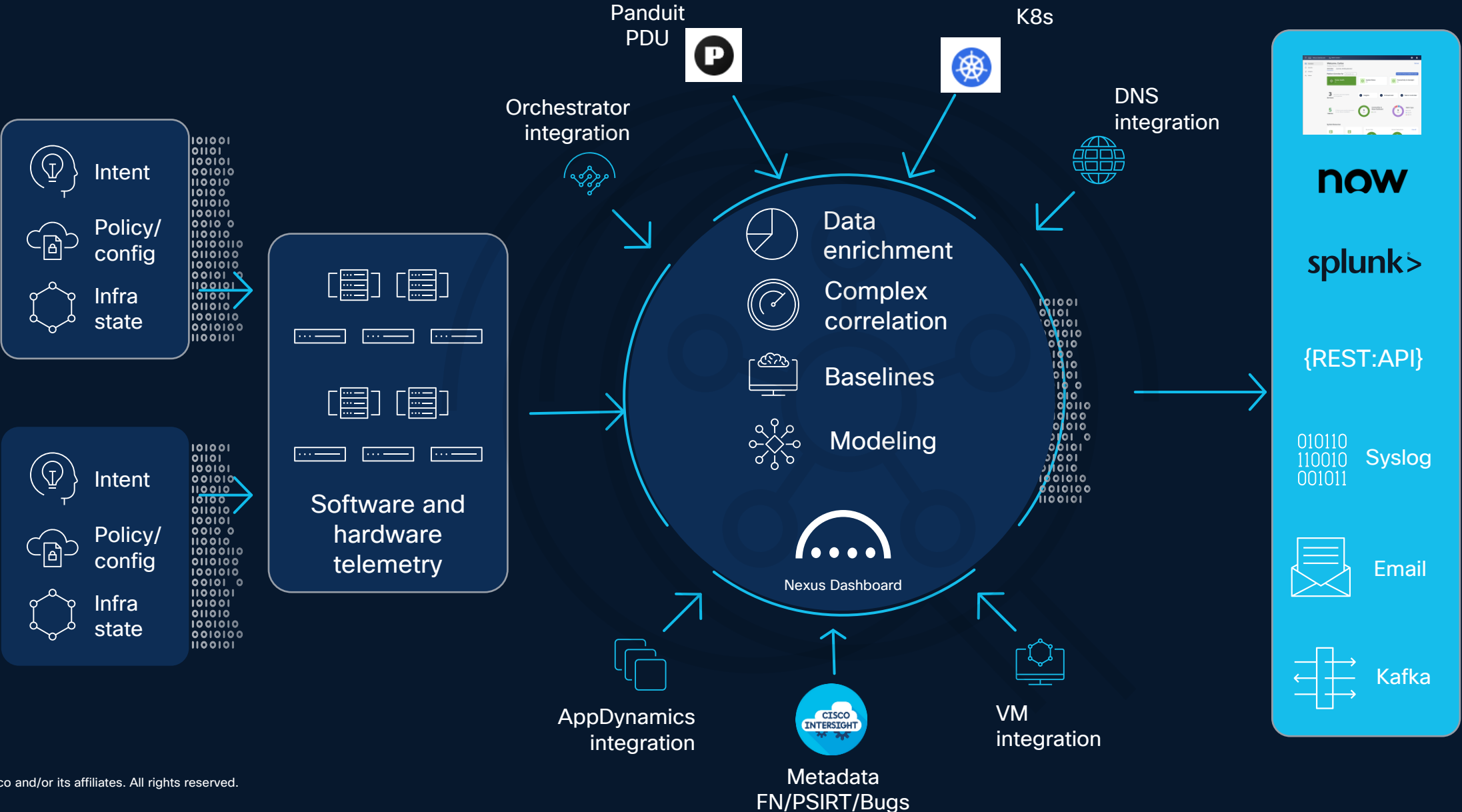
# ND Telemetry Architecture



Nexus Dashboard

Virtual form-factor

Physical form-factor

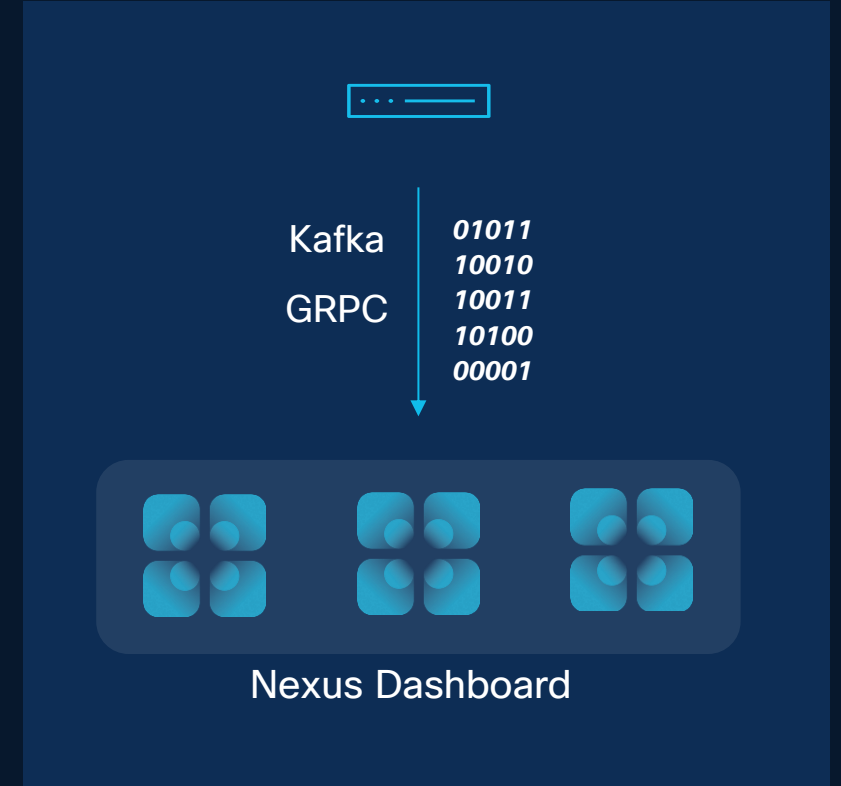




# Telemetry Data Collection

## Software Telemetry

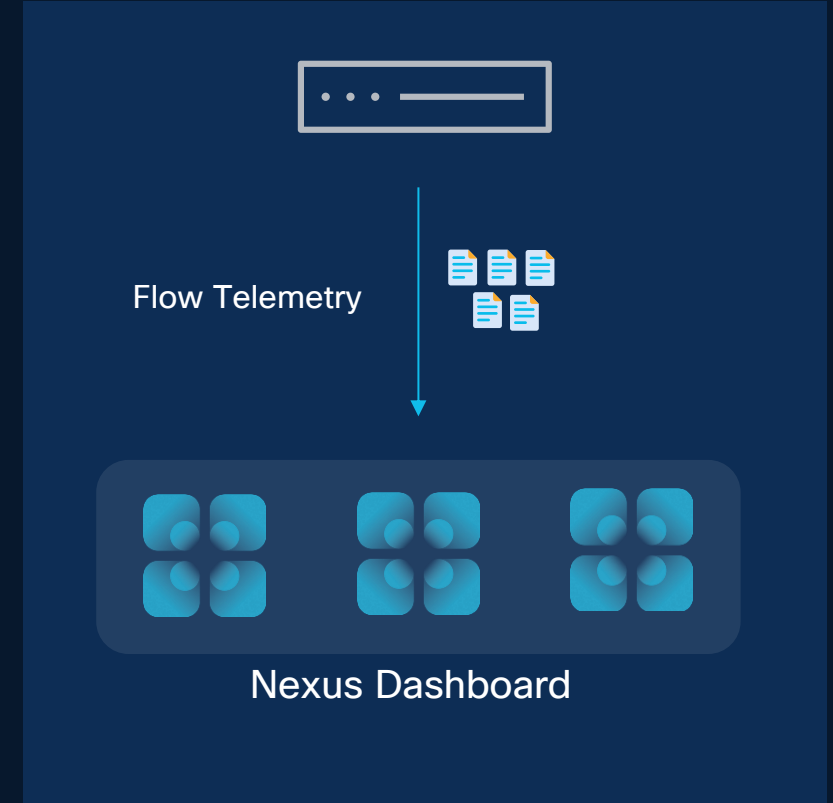
- Software Telemetry allow devices to export data only with an initial subscription request
- Telemetry can collect any objects stored in the Data Management Engine(DME), i.e.
  - Interface and hardware statistics
  - Endpoints
  - Show commands output
  - Routing protocols information
- Real Time telemetry allows on-demand subscription of specific resources for a limited time



# Telemetry Data Collection

## Hardware Telemetry

- Hardware Telemetry exports data about
  - Flow Information
    - Flow Telemetry
    - Traffic Analytics (NetflowV9 Full)
  - Packet Drops (Flow Telemetry Events)

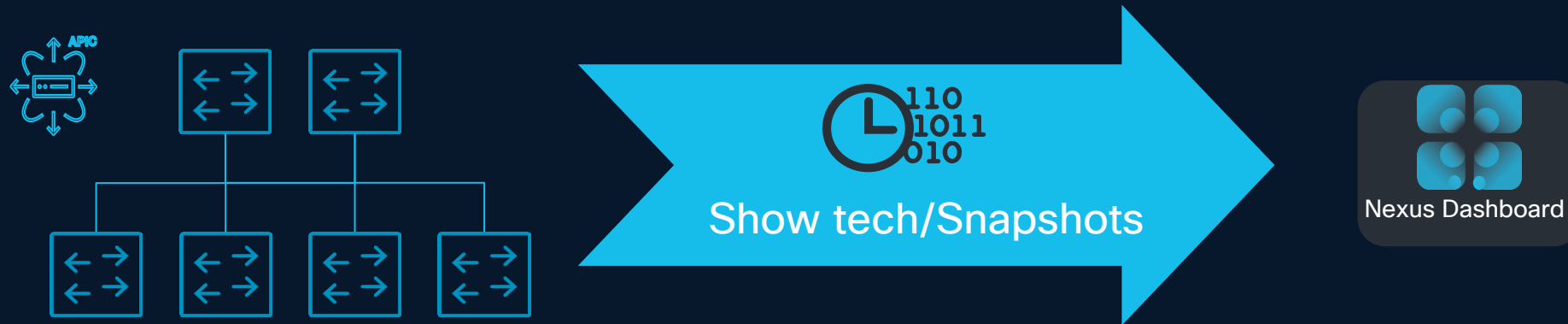


\*Traffic Analytics involves some CPU operations too

# Telemetry Data Collection

## Fabric Snapshots and Show Tech Collection

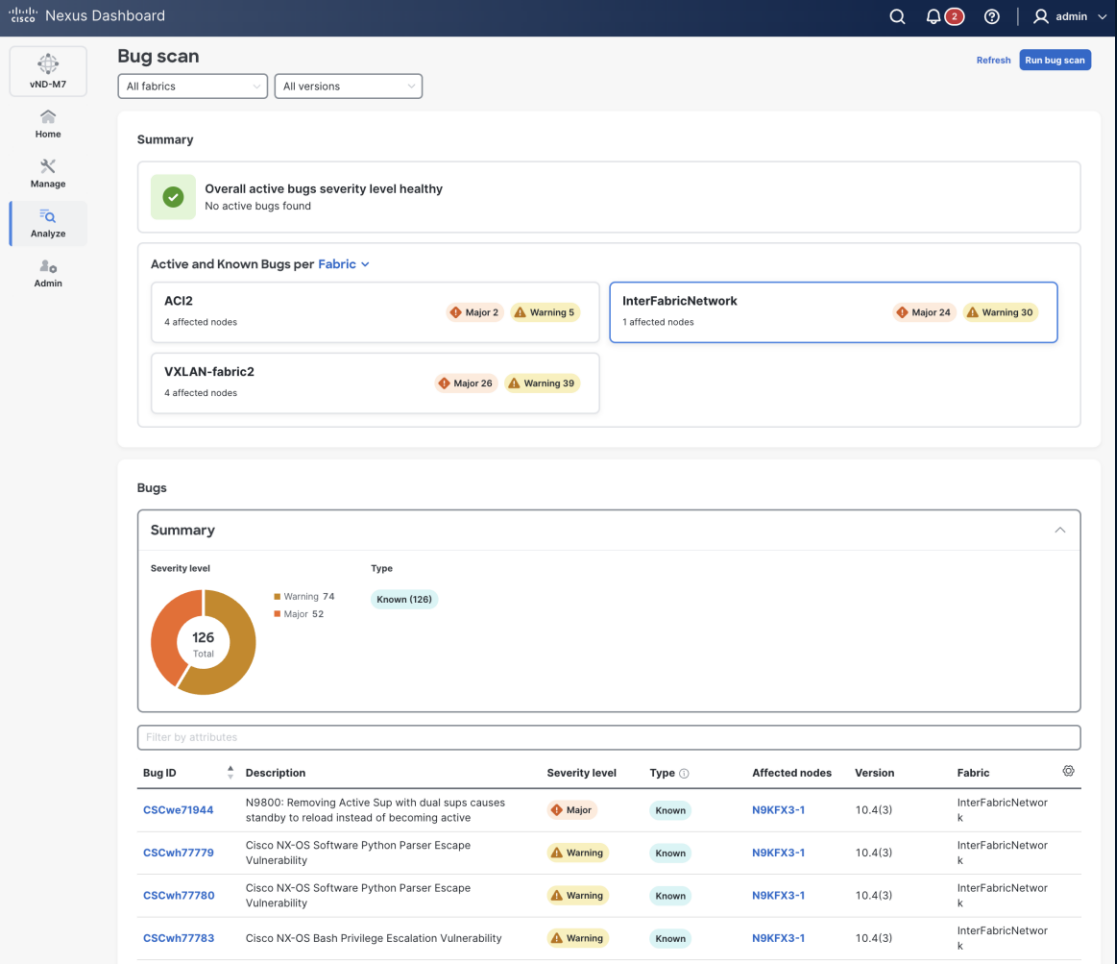
- Every 2 hours Insights performs a full snapshot of the fabric
- Every week\* show tech is collected for all switches and controllers
- Data is stored for 30+ days in historical DB



\*2 weeks if fabric has more than 100 switches

# Bug Scan

show tech and signature based



Known bugs based on Cisco RN



Active bugs through digital signature

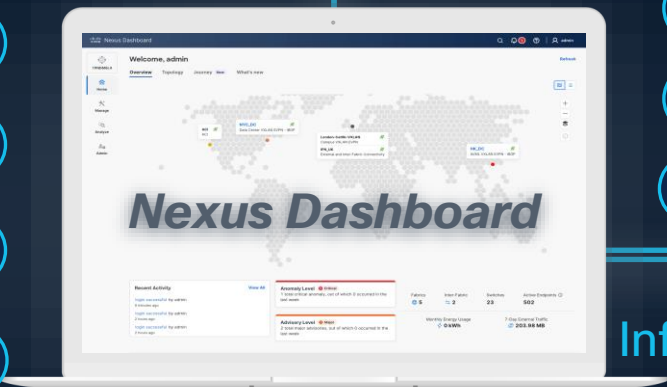


# Cisco Nexus Dashboard

AI capabilities – What is here today!

## Descriptive analysis

- Hardware conformance
- Software conformance
- Scale conformance
- Configuration compliance
- SAN Insights
- Delta analysis
- Automated bug scan
- Global Product Search
- Advisory reporting



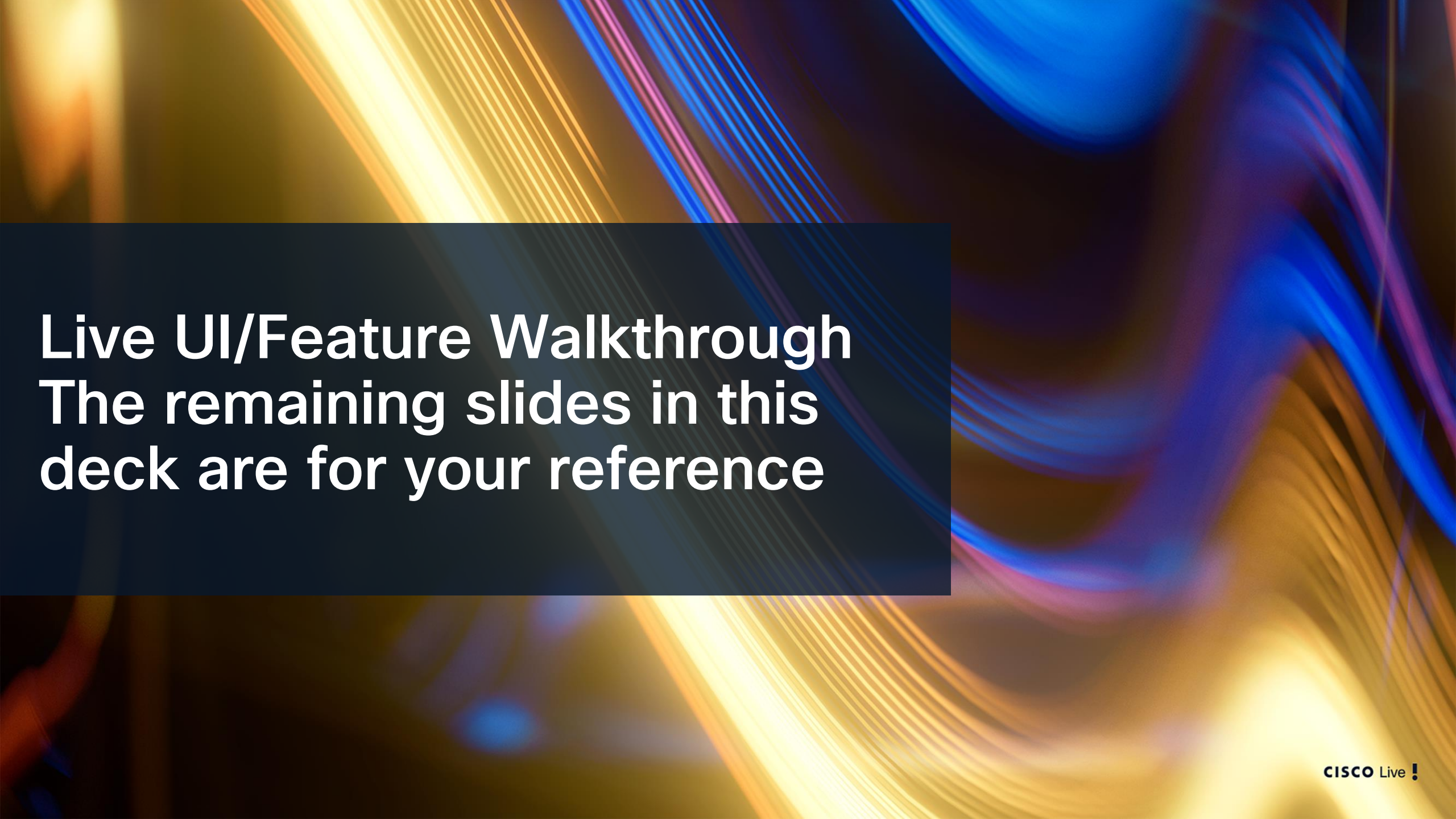
## Time series analysis

- Traffic analytics
- Interface consumption
- Energy Management analysis
- Optics (SFP) Failure Prediction
- Micro-burst detection

## Inferential/Correlation analysis

- Root cause analysis
- Anomaly correlation

# Features



**Live UI/Feature Walkthrough**  
The remaining slides in this  
deck are for your reference

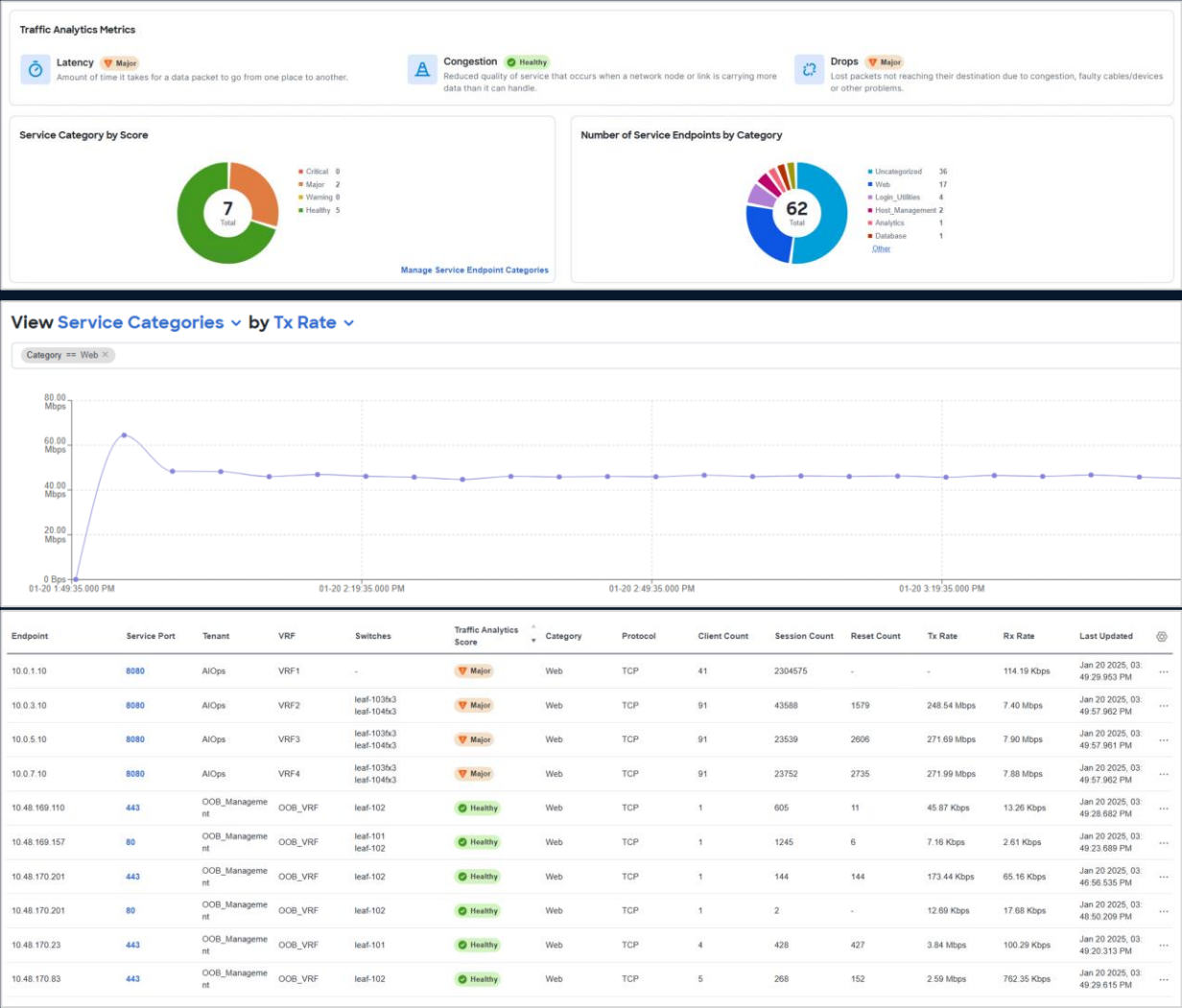


# Demos

# How to troubleshoot an application connectivity issue

# Traffic Analytics

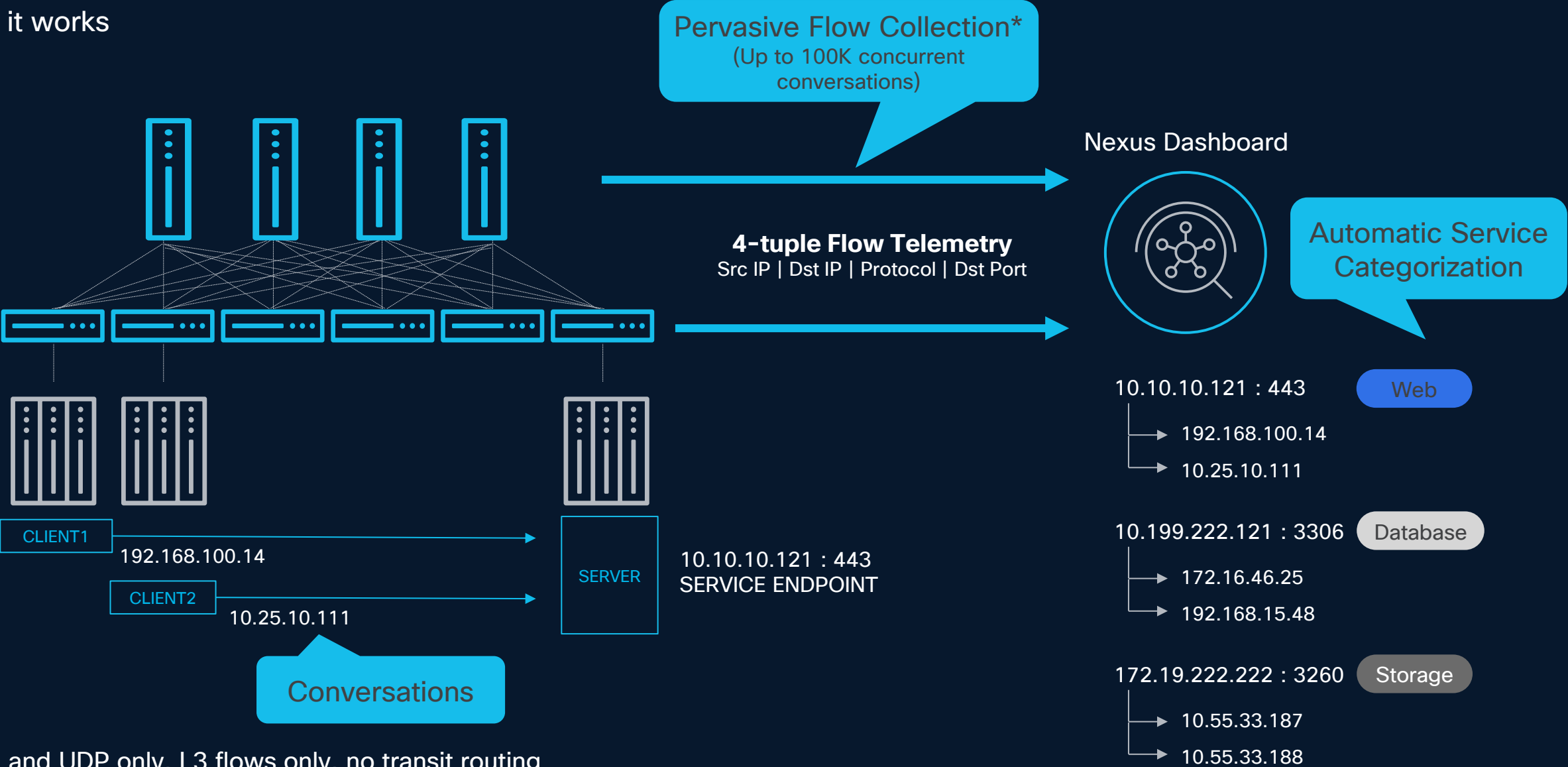
## Highlights and benefits



- Monitor all flows for Congestion, Latency, Drops
- Traffic Categorization
- Pervasive Visibility
- Detect potential issues faster
- Historical flow data maintained for up to 7 days

# Traffic Analytics

How it works



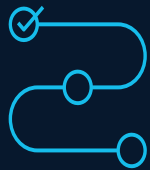
\*TCP and UDP only, L3 flows only, no transit routing



# Connectivity Analysis

# Connectivity Analysis

On Demand Helper



Traces all possible forwarding paths between source and destination



Identifies if there are problems in the path and where the issues are



Helps narrowing down the root cause, running forwarding checks

Source: Prod / AI Ops / VRF1 / IP 10.0.0.10

Destination: Prod / AI Ops / VRF1 / IP 10.0.1.100

Layer 4 Parameters

Protocol: TCP

Port Number:

Analysis Options: ☒ ELAM

Hide Job Details Re-Run Analysis

|                              |                              |                     |                         |                      |
|------------------------------|------------------------------|---------------------|-------------------------|----------------------|
| Creation Time                | End Time                     | Run Time            | Site                    | Source IP            |
| Jan 16 2025, 10:57:04.000 PM | Jan 16 2025, 10:58:40.000 PM | 1 Minute 28 Seconds | Prod                    | 10.0.0.10            |
| Destination IP               | Tenant Name                  | VRF Name            | Destination Tenant Name | Destination VRF Name |
| 10.0.1.100                   | AI Ops                       | VRF1                | AI Ops                  | VRF1                 |
|                              |                              |                     |                         | Protocol             |
|                              |                              |                     |                         | TCP                  |

Analysis Complete

PROD ACI Site

# Connectivity Analysis

## Analysis Options

### ELAM

Supported on: ACI;NX-OS

ELAM captures live traffic crossing the devices

Then a report will be generated and retrieved by ND

ELAM can provide additional information on why the packets are dropped

### OAM

Supported on: NX-OS

OAM is a protocol that facilitates troubleshooting in VXLAN EVPN

The main goal is to understand if the path between two EPs is healthy

No real traffic between EP is required.  
Can be generated by the devices

### Consistency-Checker

Supported on: NX-OS

CC is a NX-OS utility that validates the consistency between the software and hardware tables

Can be used to detect a mis programming caused by a bug or an unexpected configuration



# CICD Pipeline

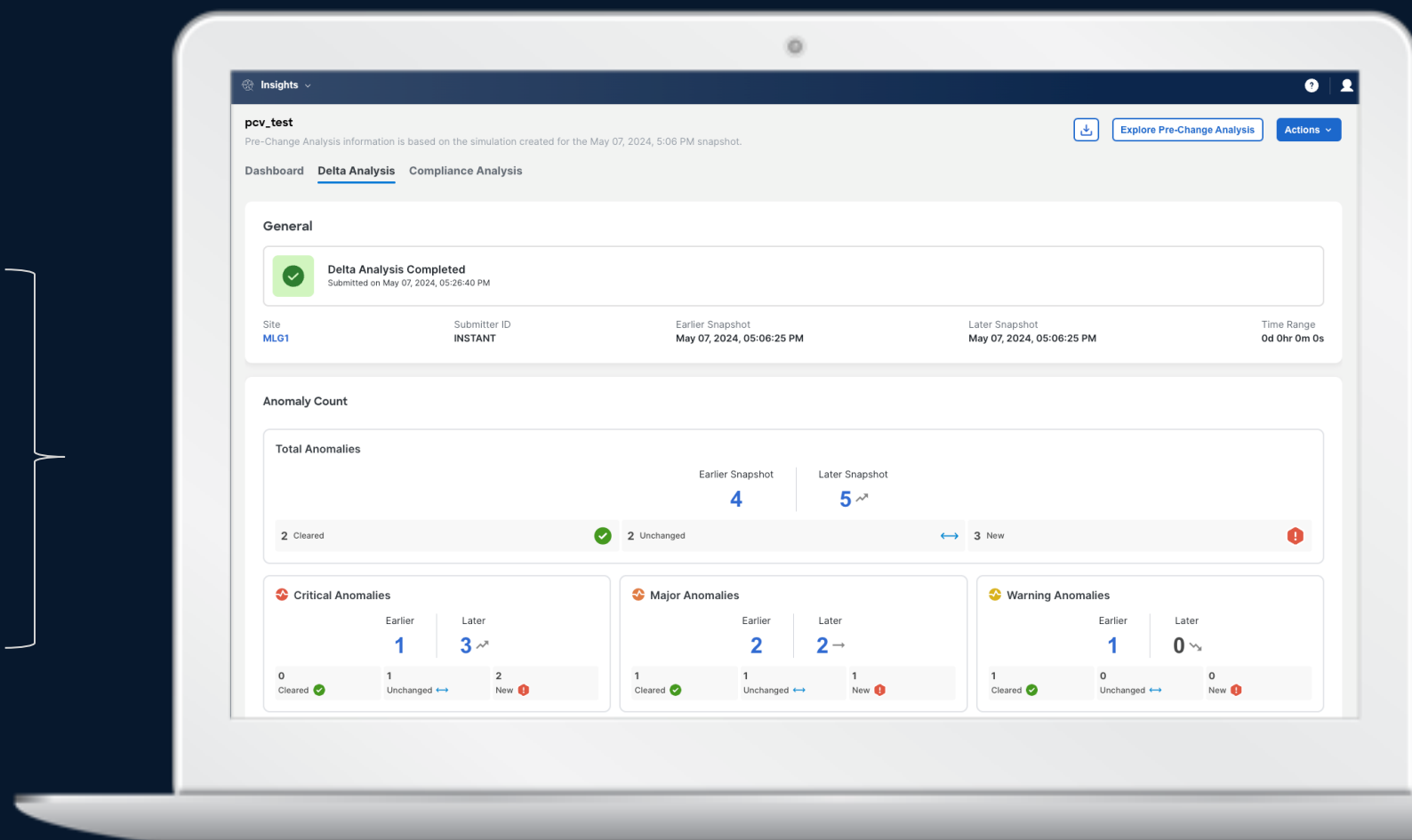
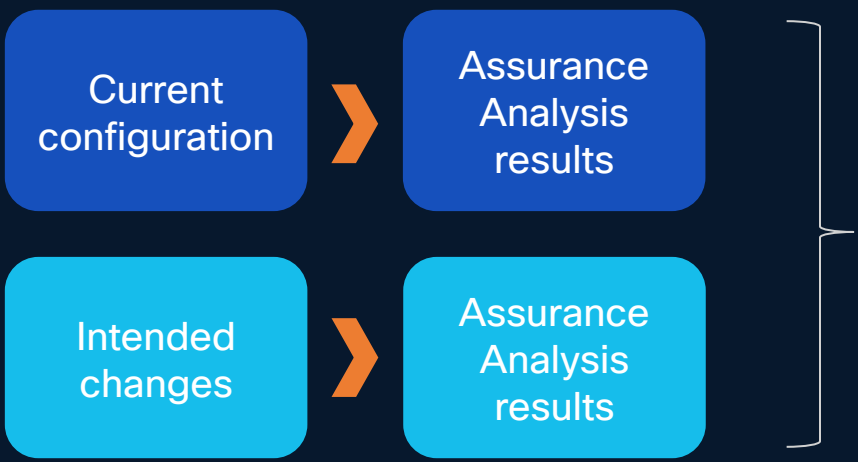
# Enhancing Change Management



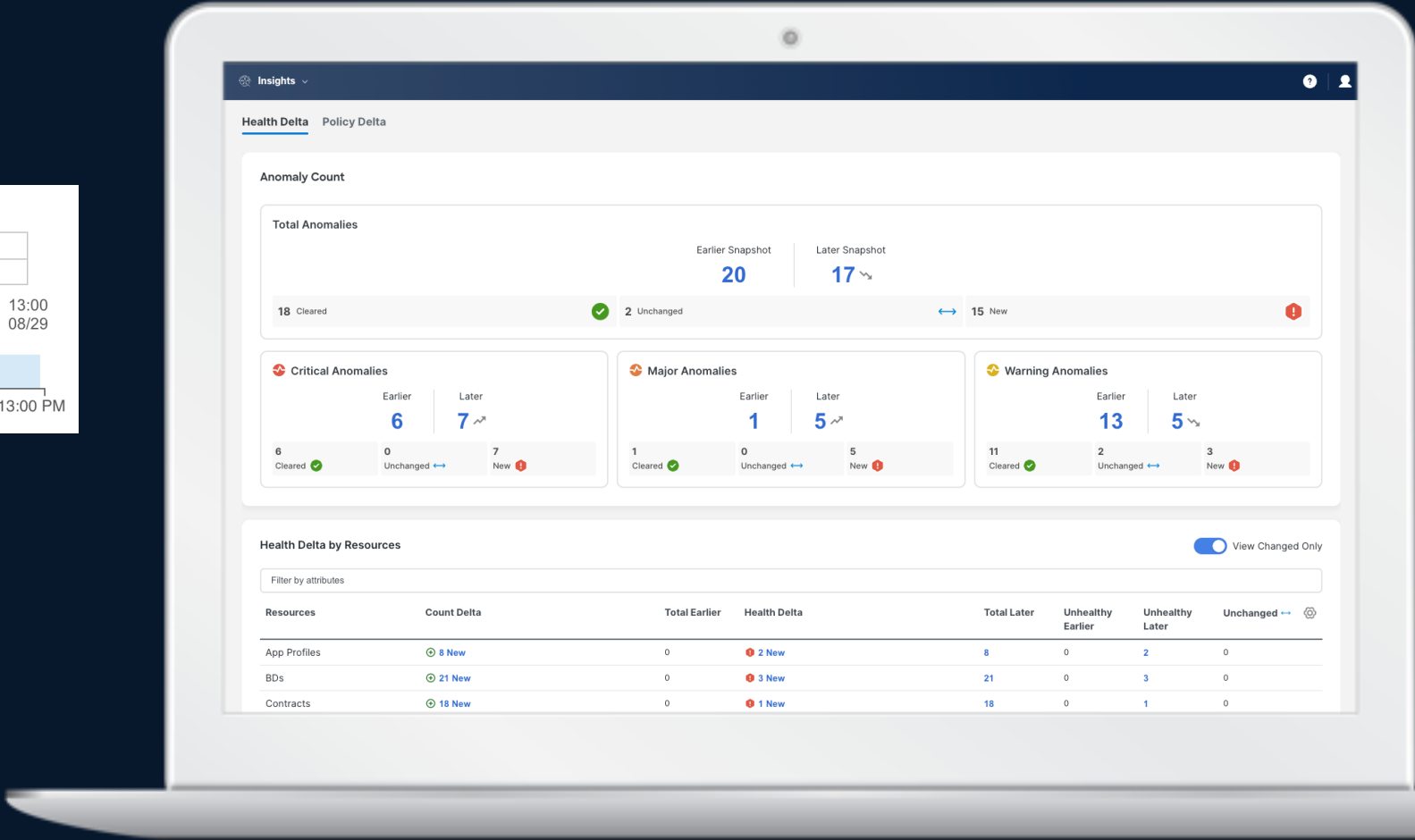
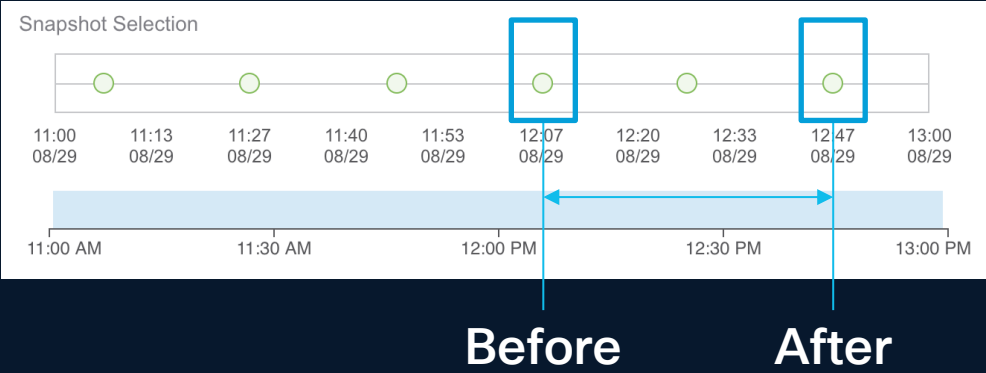
# Enhancing Change Management

See impact of intended changes

## Model intended configuration



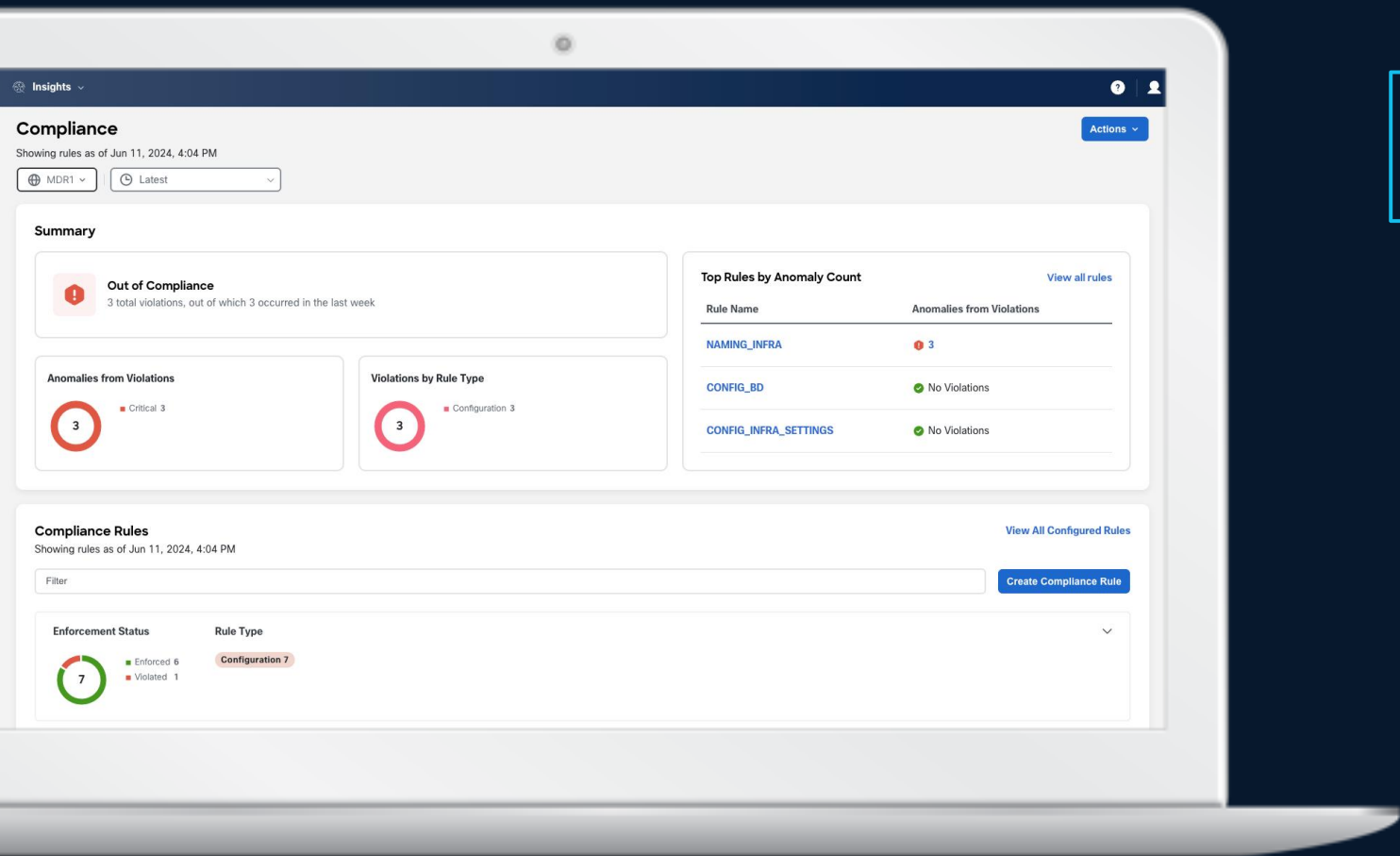
# Enhancing Change Management



# Enforcing Config Compliance

# Configuration Compliance

## Use Case Overview



**Naming Convention**  
“VRF names shall end with \_vrf”

**Configuration Rules**  
“VRF shall be enforced”



# Configuration Compliance using Insights

One option for each Use Case



## Snapshot-based

Compares the current config against the selected snapshot, allowing or not the addition of new objects



## Import-based

Compares the current config against a partial config, allowing or not the addition of new objects



## Template-based

Verifies that the attributes of a set of selected objects match the criteria specified in a template

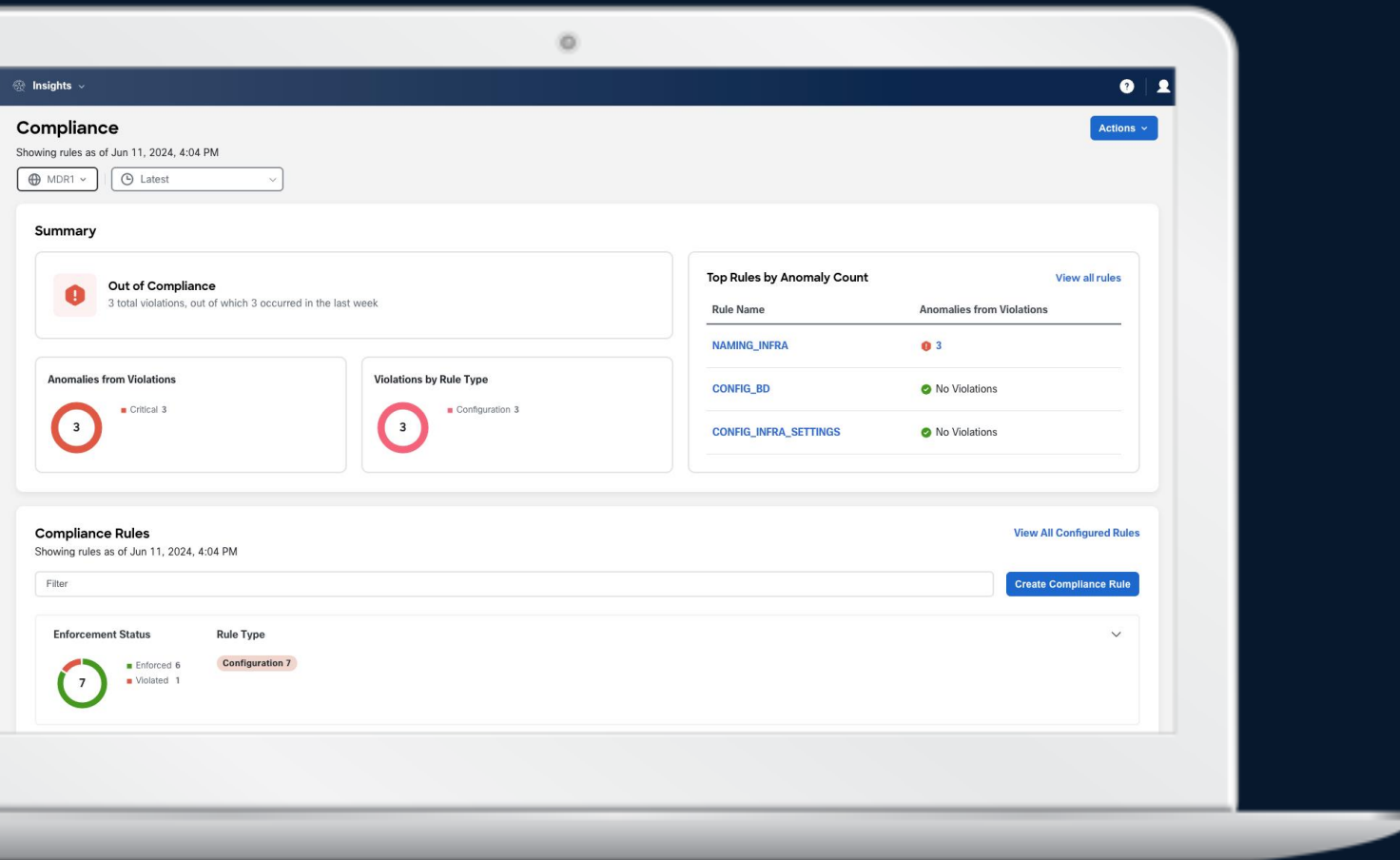


## Manual

Verifies that the attributes of a set of selected objects match the criteria specified manually in the UI

# Configuration and Segmentation Compliance

## Use Case Overview



## Segmentation

“PCI must not talk to non PCI”

## SLA

“APP must talk to DB on 1433”

## Traffic Restriction

“INET must not talk to WEB on ssh”



# Firmware Update Analysis

# Upgrading with Confidence

## Typical Challenges



Assurance that all pre-requisites have been satisfied



Time required to run all pre-checks and post-checks



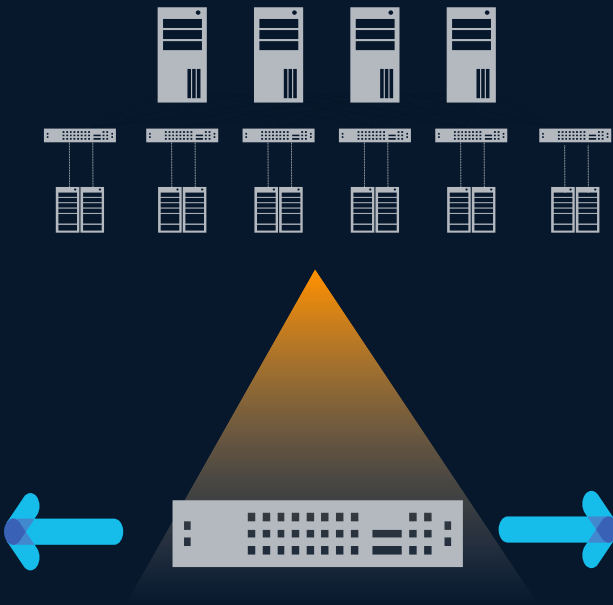
Complexity of analysis of potential issues in target release



Lack of visibility of potential impact

# Upgrading with Confidence

Pre-Upgrade and Post-Upgrade



## Pre-upgrade checks

|                      |   |
|----------------------|---|
| Physical interfaces  | ✓ |
| Neighbors            | ✓ |
| Routes               | ✓ |
| Endpoints            | ✓ |
| Capacity utilization | ✓ |

## Post-upgrade checks

|                                     |   |
|-------------------------------------|---|
| Physical interfaces                 | ✓ |
| Neighbors                           | ✓ |
| Routes                              | ✓ |
| Endpoints (lower number discovered) | ✗ |
| Capacity utilization                | ✓ |

# Upgrading with Confidence

Pre-Upgrade and Post-Upgrade

- Pre-Upgrade Checks
- Version and Bug Info / Validation
- Advice about potential conditions that exist before upgrade (not just faults)
- Post-Upgrade Delta Analysis
  - *Did I fix it? Did I break it? Did I change it?*

The image shows a laptop screen displaying the Cisco Insights web interface. The interface is divided into three main sections: Physical Interfaces, VPC Interfaces, and BGP Protocol. Each section contains a table comparing 'Earlier' and 'Later' states for various metrics.

**Physical Interfaces**

| Node     | Interface | Type     | Admin State |       | Operation |
|----------|-----------|----------|-------------|-------|-----------|
|          |           |          | Earlier     | Later | Earlier   |
| Leaf-602 | eth1/4    | physical | Up          | Down  | Up        |
| Leaf-602 | eth1/1    | physical | Up          | Down  | Up        |

10 Rows

**VPC Interfaces**

| Node     | Interface | Type | Admin State |       | Local Operational State |       |
|----------|-----------|------|-------------|-------|-------------------------|-------|
|          |           |      | Earlier     | Later | Earlier                 | Later |
| Leaf-602 | pod1-vmm  | vpc  | Up          | Up    | Up                      | Down  |

10 Rows

**BGP Protocol**

| Node     | Number of Configured Interfaces |       | Neighbor Count |       |
|----------|---------------------------------|-------|----------------|-------|
|          | Earlier                         | Later | Earlier        | Later |
| Leaf-602 | 2                               | 1     | 2              | 1     |

10 Rows



# Conclusion

# Conclusion

- Nexus Dashboard is the most comprehensive tool for operating Nexus 9000 fabrics. It can help with and not only:
  - Assurance
  - Conformance
  - Monitoring with Real Time Telemetry
  - Reporting
  - Compliance
  - Software and Change Management
- For highly customizable dashboard and basic inventory and statistics telemetry visibility, opensource tools are available and can combine with ND capabilities

# Complete your session surveys



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(from 11:30 on Thursday, while supplies last)

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**Visit** the Technical Solutions Clinics to discuss your technical questions



**Attend** the interactive education with DevNet, Capture the Flag, and Walk-in Labs



**Visit** the On-Demand Library for more sessions at [CiscoLive.com/On-Demand](https://CiscoLive.com/On-Demand)



Thank you

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